

Sharmin Afrose

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Professional Experience

Oak Ridge National Laboratory

Oak Ridge, TN

Research and Development Staff

Mar. 2026 - Present

- **Agentic AI on Quantum Computing:** Leading the Quantum Science Center's (QSC) AI-for-quantum software work. Developing ChatQEC, which grounds every answer in cited literature so users can verify it, and OrcheQ, a multi-agent system for quantum circuit code generation, optimization, and quantum error correction.
- **Suicide Predictive Modeling:** Developed a subgroup-aware risk prediction method that keeps the model interpretable, so clinicians can act on the predictions, for the US Department of Veterans Affairs (VA) population.
- **Distributed Quantum Computing:** Built an agentic MLIR-to-QIR translation pipeline with dual-backend verification and an end-to-end quantum circuit cutting pipeline. Releasing two tools under MACH-Q: Agentic-MLIR-to-QIR and Qweaver/mlir-cut.

Oak Ridge National Laboratory

Oak Ridge, TN

Postdoctoral Research Associate

Mar. 2023 - Feb. 2026

- Focusing on predictive machine learning models and biostatistical methods to identify high-risk individuals for suicide using US Department of Veterans Affairs (VA) data, ensuring model effectiveness and fairness across subgroups. Quantification and mitigation of bias on whole US patient-level studies.
- Designing a cross-platform Quantum Intermediate Representation Execution Engine (QIR-EE) to enable hardware-agnostic quantum workflows. Explored MLIR-based optimizations to enhance quantum program compilation and execution.
- Developing a high-performance, privacy-preserving federated learning (FL) framework for scientific applications, funded by a SEED proposal award. Utilized HPC resource (Frontier Supercomputer) to utilize distributed parallel training on multiple GPU nodes for multiple clients in FL setup. Investigated sketching techniques to enhance scalability and communication efficiency in distributed FL environments.

Virginia Tech

Blacksburg, VA

Graduate Research Assistant

Aug. 2018 - Dec. 2022

- Analyzed bias among different demographic subgroups (e.g., ethnicity, age group, and gender) for imbalanced MIMIC III and SEER clinical datasets
- Analyzed bias in different prediction tasks (e.g., in-hospital mortality prediction, decompensation prediction using MIMIC III dataset, and breast and lung cancer survivability prediction using SEER dataset)
- Designed a double prioritized (DP) technique to reduce the bias and improve prediction accuracy among minority subgroups (e.g., Asian, African American, and older populations)
- Developed CryptoAPI-Bench benchmark to identify the limitations of Java cryptographic vulnerability detection tools. Parfait (crypto vulnerability detection tool by Oracle) is also used our benchmark to improve and compare their tools
- Contributed to improving CryptoGurad (a static analysis tool to detect cryptographic vulnerability in Java program)

Oak Ridge National Laboratory

Oak Ridge, TN

ORISE Summer Intern

Jun. 2021 - Aug. 2021

- Comparative analysis of different LDP frameworks including distribution-based approaches, randomized response-based approaches, count sketch-based approaches for IoT-based streaming data

Virginia Tech

Blacksburg, VA

Graduate Teaching Assistant

Spring 2020

- Assisted with teaching, grading on the Course "Principle Computer Security"

BRAC University

Dhaka, Bangladesh

Lecturer in the department of Computer Science and Engineering

Sep. 2015 - Jul. 2018

- Conducted several courses (e.g., Operating System, Programming Language, Data Structure) and mentored students
- Research on measuring the reliability or correctness of the Twitter post using Natural Language Processing (NLP)

- Developed a secret sharing approach to compute the probability of a patient developing diseases without revealing sensitive data
- Designed a system to mine frequent itemsets using local differential privacy without revealing personal sensitive itemsets

Education

Virginia Tech

Blacksburg, VA

PhD in Computer Science and Application (GPA: 4.00/4.00)

Aug. 2018 - Dec. 2022

- Dissertation Title: Methodology Development for Improvement of Critical Classification Applications
- Supervisor: Danfeng (Daphne) Yao, Ph.D. (Elizabeth and James E. Turner Jr. '56 Faculty Fellow)
- Dissertation Committee Members
 - T. M. Murali (Virginia Tech), Na Meng (Virginia Tech), B. Aditya Prakash (Georgia Tech), Sharon Xiaolei Huang, (Penn State University)

Bangladesh University of Engineering and Technology

Dhaka, Bangladesh

BS in Computer Science and Engineering (GPA: 3.89/4.00)

2010-2015

Technical Skills

Programming Python, SQL, Java, C, C++, MATLAB, Assembly, RUST, Julia, R, PHP

Machine Learning (XGBoost, Logistic Regression, Support Vector Machine, Decision Tree, Random

AI/ML Forest, Ensemble, Anomaly detection), Deep Learning (TabNet, CNN, MLP, LSTM, GAN, Transformer, NLP), TensorFlow, PyTorch, Keras and other libraries**Generative AI Tools** GitHub Copilot, ChatGPT, Liner, Adobe AI Assistant

Publications

1. **Sharmin Afrose**, Vicente Leyton-Ortega, Narasinga Rao Miniskar, Elaine Wong, and Travis S. Humble (2026). Agentic MLIR-to-QIR Translation with Dual-Backend Testing: A Hybrid Deterministic and LLM Approach. *Agenti-CAI4HPC'26 (Workshop at SC'26)* (accepted)
2. **Sharmin Afrose**, Ethan Seefried, Nicholas Stapleton, Vicente Leyton Ortega, Travis S. Humble, and Tirthankar Ghosal (2026). LLM-Based Quantum Coding Assistants: A Survey of Approaches, Benchmarks, and Open Challenges. *IEEE Quantum Week (QCE'26)* (accepted)
3. **Sharmin Afrose**, William H. Kazanis, Jodie Trafton, Anuj Kapadia, and Alina Peluso (2026). Subgroup-aware Suicide Risk Prediction via Functionally Adaptive Interaction Regularized Regression. *npj Mental Health Research*. doi:10.1038/s44184-026-00229-y
4. Nicholas Stapleton, **Sharmin Afrose**, Ethan Seefried, Vicente Leyton Ortega, Travis S. Humble, and Tirthankar Ghosal (2026). Quantum-RAG: Retrieval-Augmented Generation Framework for Quantum Circuit Code Generation. In *2026 IEEE 19th Dallas Circuits and Systems Conference (DCAS)*, pp. 1--5. IEEE. doi:10.1109/DCAS69364.2026.11544968
5. **Sharmin Afrose**, Snehalatha Reddy Eggina, Jorge Ramirez Osorio, Anuj Kapadia, Jodie Trafton, and Alina Peluso (2025). Algorithmic Bias in Suicide Risk Prediction Models. In *Generative AI in Healthcare: Transforming Diagnostics, Treatment, and Patient Outcomes*, pp. 185--200. Springer Nature Switzerland. doi:10.1007/978-3-032-11999-5_14
6. **Sharmin Afrose**, Olivera Kotevska (2025). Dynamical Sketching for Enhanced Communication Efficiency in Federated Learning. In *2025 IEEE Conference on Artificial Intelligence (CAI)*, pp. 982--987. IEEE. doi:10.1109/CAI64502.2025.00172
7. Elaine Wong, Vicente Leyton-Ortega, Daniel Claudino, Seth R. Johnson, Austin J. Adams, **Sharmin Afrose**, Meenambika Gowrishankar, Anthony Cabrera, and Travis S. Humble (2025). A Cross-platform Execution Engine for the Quantum Intermediate Representation. *The Journal of Supercomputing*, 81(16):1521. doi:10.1007/s11227-025-07969-2
8. Tanmoy Sarkar Pias, **Sharmin Afrose**, Moon Das Tuli, Ipsita Hamid Trisha, Xinwei Deng, Charles B. Nemeroff and Danfeng Yao (2025). Low Responsiveness of Machine Learning Models to Critical or Deteriorating Health Conditions. *Communications Medicine*, 5, 62. doi: 10.1038/s43856-025-00775-0

9. Vicente Leyton Ortega, Daniel Claudino, Travis Humble, Elaine Wong, **Sharmin Afrose**, Meenambika Gowrishankar, and Seth Johnson (2024). XACC Quantum Execution Engine: Streamlining Quantum Computing through QIR Interpretation. *Bulletin of the American Physical Society*
10. Elaine Wong, **Sharmin Afrose**, Meenambika Gowrishankar, Daniel Claudino, Vicente Leyton-Ortega, Seth Johnson and Travis Humble. A QIR Toolchain with XACC. *In 2023 IEEE International Conference on Quantum Computing and Engineering (QCE) (Vol. 2, pp. 205-206). IEEE. 2023*
11. **Sharmin Afrose**, Wenjia Song, Charles B. Nemeroff, Chang Lu, Danfeng (Daphne) Yao (2022). Subpopulation-specific Machine Learning Prognosis for Underrepresented Patients with Double Prioritized Bias Correction. *Nature Communications Medicine*, 2, 111. doi: 10.1038/s43856-022-00165-w
12. Danfeng (Daphne) Yao, Sazzadur Rahaman, Ya Xiao, **Sharmin Afrose**, Miles Frantz, et al. (2022). Being the Developers' Friend: Our Experience Developing a High-Precision Tool for Secure Coding. *IEEE Security & Privacy*, pp. 2--11. doi: 10.1109/MSEC.2022.3159481
13. **Sharmin Afrose**, Ya Xiao, Sazzadur Rahaman, Barton P. Miller, Danfeng (Daphne) Yao (2022). Development of Benchmarks for Java Cryptographic APIs and Evaluation of Static Vulnerability Detection Tools. *IEEE Transactions on Software Engineering*, pp. 1-1, 5555. doi:10.1109/TSE.2022.3154717
14. **Sharmin Afrose**, Danfeng (Daphne) Yao, & Olivera Kotevska (2021). Measurement of Local Differential Privacy Techniques for IoT-based Streaming Data. *18th International Conference on Privacy, Security and Trust (PST)*, pp. 1--10. IEEE. doi:10.1109/PST52912.2021.9647839
15. **Sharmin Afrose**, Tanzima Hashem, and Mohammed Eunus Ali (2021). Frequent Itemsets Mining with a Guaranteed Local Differential Privacy in Small Datasets. *In 33rd International Conference on Scientific and Statistical Database Management (SSDBM)*, pp. 232--236. doi:10.1145/3468791.3468807
16. Sazzadur Rahaman, Ya Xiao, **Sharmin Afrose**, Ke Tian, Miles Frantz, Na Meng, Barton P. Miller, Fahad Shaon, Murat Kantarcioglu, and Danfeng (Daphne) Yao (2020). Deployment-quality and Accessible Solutions for Cryptography Code Development. *In Proceedings of the Tenth ACM Conference on Data and Application Security and Privacy*, pp. 174--176. doi:10.1145/3374664.3379536
17. **Sharmin Afrose**, Sazzadur Rahaman, & Danfeng (Daphne) Yao (2019). CryptoAPI-Bench: A Comprehensive Benchmark on Java Cryptographic API Misuses. *IEEE Secure Development Conference (SecDev)*, pp. 49--61. IEEE. doi:10.1109/SecDev.2019.00017
18. Sazzadur Rahaman, Y. Xiao, **Sharmin Afrose**, Fahad Shaon, Ke Tian, Miles Frantz, Murat Kantarcioglu, Danfeng (Daphne) Yao (2019). CryptoGuard: High Precision Detection of Cryptographic Vulnerabilities in Massive-sized Java Projects. *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pp. 2455--2472. doi:10.1145/3319535.3345659
19. S. T. Al-Amin, S. Tairin, **Sharmin Afrose**, W. Mohammad and M. Naznin (2018). Semantic Clustering in Credible Human Sensed Event Detection, 14th International Conference on Distributed Computing in Sensor Systems (DCOSS), New York, NY, USA, 2018, pp. 119-120, doi:10.1109/DCOSS.2018.00026
20. Nusrat Jahan Mozumder, Maitraye Das, Tanzima Hashem, **Sharmin Afrose**, and Khandakar Ashrafi Akbar (2019). Towards Privacy-preserving Authenticated Disease Risk Queries. *Journal of Information Processing*, 27:624--642. doi:10.2197/ipsjip.27.624
21. Maitraye Das, Nusrat Jahan Mozumder, **Sharmin Afrose**, Khandakar Ashrafi Akbar, Tanzima Hashem (2018). A Novel Secret Sharing Approach for Privacy-Preserving Authenticated Disease Risk Queries in Genomic Databases. *Annual Computer Software and Applications Conference (COMPSAC)*, vol. 1, pp. 645--654. IEEE. doi:10.1109/COMPSAC.2018.00097
22. Syed Ishtiaque Ahmed, Steven J Jackson, Maruf Zaber, Mehrab B. Morshed, Md Habibullah Bin Ismail, and **Sharmin Afrose** (2013). Ecologies of Use and Design: Individual and Social Practices of Mobile Phone Use Within Low-literate Rickshawpuller Communities in Urban Bangladesh. *In Proceedings of the 4th Annual Symposium on Computing for Development*, pp. 1--10. doi:10.1145/2537052.2537066

Presentations (Selected)

1. *Life Within a Lab and a NQIS Research Center. ASCR QIS PI Meeting*, Rockville, MD.
2. AI for Quantum at the Quantum Science Center: Towards Building a Connective Layer Across the Quantum Computing Stack. *QCUF'26*.
3. OrcheQ: Multi-Agent Quantum Code Orchestration. *QSC All Hands Meeting'26*, Gatlinburg, TN.
4. Connecting AI Agents for Scientific Analysis and Knowledge-Base Curation. *OSDX'26*, Oak Ridge, TN.
5. **Sharmin Afrose**, Snehalatha Reddy Eggina, Jorge Ramirez Osorio, Anuj Kapadia, Jodie Trafton, Alina Peluso. Algo-

- rithmic Bias in Suicide Risk Prediction Models. *Machine Learning for Health Symposium*, Vancouver, Canada, December 2024.
6. Olivera Kotevska, Ryan Prout, **Sharmin Afrose**, Alan Longcoy. Federated Learning in Heterogeneous Environments for Science Applications, *Oak Ridge National Laboratory's Core Universities AI Workshop*. Raleigh, North Carolina, October 2024.
 7. Snehalatha Reddy Eggina, **Sharmin Afrose**, Alina Peluso. Addressing Algorithmic Bias in Multi-Class Predictive Modeling of Suicide, *ORNL Summer Student Symposium*, Oak Ridge, Tennessee, August 2024.
 8. Daniel Li, **Sharmin Afrose**, Olivera Kotevska. Exploring Communication Efficiency in Federated Learning, *DOE Cybersecurity and Technology Innovation Conference*, Dallas, Texas, July 2024.
 9. Elaine Wong, Vicente Leyton-Ortega, Daniel Chaves-Claudino, Meena Gowrishankar, Seth Johnson, **Sharmin Afrose**, Travis Humble. Making QIR Executable, *Fourth International Workshop on Quantum Computing Software (adjunct with Supercomputing)*, Denver, Colorado, November 2023.
 10. **Sharmin Afrose**, Peter Groszkowski. *ORNL Software and Data Expo*, Oak Ridge, TN, September 2023.
 11. **Sharmin Afrose**, Danfeng Yao, Olivera Kotevska, Measurement of Local Differential Privacy Techniques for IoT-based Streaming Data, *International Conference on Privacy, Security and Trust (PST)*, New Zealand (Virtual), December 2021.
 12. **Sharmin Afrose**, Sazzadur Rahaman, & Danfeng (Daphne) Yao (2019). CryptoAPI-Bench: A Comprehensive Benchmark on Java Cryptographic API Misuses. *IEEE Secure Development Conference (SecDev)*, McLean, VA, September 2019.

Software Releases

- **ChatQEC**: <https://code.ornl.gov/qsc-as/chatqec>
- **Agentic-MLIR-to-QIR**: https://code.ornl.gov/mach-q/quantum-mlir-taxonomy/agentic_mlir_qir
- **Qweaver**: <https://code.ornl.gov/mach-q/quantum-mlir-taxonomy/mlir-cut>

Honors and Awards

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| 2026 | Distinguished Synergy Award (ORNL Quantum Team) , IEEE Quantum Technical Community (QTC) |
| 2026 | Winning Team, Explainable Suicide Risk Detection Challenge , IEEE Big Data 2026 Challenge |
| 2023 | Best Presentation Award , ORNL Software and Data Expo (OSDX) |
| 2022 | Participant, Scholarship from Association for Women in Computing (AWC) , Grace Hopper Celebration (GHC) |
| 2021 | Participant , Selected from CRA-W Grad Cohort Workshop for Women |
| 2019 | Participant , Travel grant from CRA-W Grad Cohort Workshop for Women |
| 2015 | Undergrad Thesis Poster Award , Bangladesh University of Engineering and Technology |
| 2011-15 | Dean's List Award , Academic excellence in Bangladesh University of Engineering and Technology |
| 2011-15 | Merit Scholarship , Academic excellence in Bangladesh University of Engineering and Technology |

Mentoring

Mentored/co-mentored 3 summer students, all reaching peer-reviewed submissions

Summer 2026

- **Nirajan Banjade** (Texas State University), PCIP'26: Quantum LLM-Wiki
- **Romit Ghosh** (Rutgers University), SULI'26: Hybrid Quantum-Classical Benchmarking
- **Sangram Deshpande** (NC State University), PQCIP'26: Quantum Hardware-aware Circuit Mapping

Volunteer Work

NQISRC, IEEE Quantum Week

Toronto, Canada

Volunteer

Sept. 2026

- Volunteered at the NQISRC booth at the IEEE Quantum Week conference.

Annoor Academy of Knoxville

Knoxville, TN

Judge, Middle School STEM Fair

Jan. 2026

- Volunteered as a judge for the middle school STEM fair, building early interest in STEM careers.